

Mohammad Zhalechian

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APPOINTMENT

Indiana University, Bloomington, IN

Assistant Professor of Operations and Decision Technologies
Kelley School of Business

2023 – Present

Harvard University, Cambridge, MA

Postdoctoral Fellow
Harvard Kennedy School

2022 – 2023

EDUCATION

University of Michigan, Ann Arbor, MI

Ph.D. in Industrial and Operations Engineering (Operations Research)
M.A. in Statistics

2022

2019

University of Tehran, Tehran, Iran

M.Sc. in Industrial Engineering and Operations Research

2017

K. N. Toosi University of Technology (KNTU), Tehran, Iran

B.Sc. in Industrial Engineering and Operations Research

2014

RESEARCH INTERESTS

- Sequential and Data-Driven Decision-Making
- Design and Evaluation of Human-AI Systems
- Algorithm Applications in Healthcare and High-Stakes Settings

PUBLICATIONS

[† industry collaborators; (M.D.) M.D. collaborators]

Published Papers

[1] Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy

Forthcoming at *Management Science*

M. Zhalechian, S. Saghaian, O. Robles[†]

- *Media coverage:* Transforming FDA Clearance: How AI and Human Insight Can Improve Medical Device Safety
- *Media coverage:* A human-algorithm approach to approving medical devices could slash the recall rate and save billions

[2] Contextual Learning with Online Convex Optimization with Applications to Medical Decision-Making

Management Science, 2025

Collaboration with Massachusetts Hospital

E. Keyvanshokoo, M. Zhalechian, C. Shi, MP. Van Oyen, P. Kazemian

- Finalist, MIF Paper Competition, 2026
- Finalist, CHOM Best Paper Competition (POMS), 2022
- Finalist, INFORMS MSOM Best Student Paper Competition, 2021
- Finalist, INFORMS Health Applications Society (HAS) Best Student Paper Competition, 2021
- Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper, 2020

[3] Dynamic Care Unit Placements under Unknown Demand with Learning

Manufacturing & Service Operations Management, 2025

Collaboration with St. Joseph Hospital

A. Dean, M. Zhalechian, MP. Van Oyen

[4] Data-driven Hospital Admission Control: A Learning Approach

Operations Research, 2023

Collaboration with St. Joseph Hospital

M. Zhalechian, E. Keyvanshokoo, C. Shi, MP. Van Oyen

- Winner, IOE Richard C. Wilson Best Student Paper Award on Service Systems, 2021
- Finalist, Seth Bonder Scholarship, INFORMS Health Applications Society, 2020

[5] Online Resource Allocation with Personalized Learning

Operations Research, 2022

Collaboration with Michigan Medicine

M. Zhalechian, E. Keyvanshokoo, C. Shi, MP. Van Oyen

[6] Validation of a Visual Field Prediction Tool for Glaucoma: A Multicenter Study Involving Patients with Glaucoma from Throughout the United Kingdom

American Journal of Ophthalmology, 2025

Collaboration with Medisoft UK

A. Dean, DJ. Fu (M.D.), M. Zhalechian, MP. Van Oyen, M. Lavieri, AP. Khawaja (M.D.), J. Stein (M.D.)

[7] Augmenting Kalman Filter Machine Learning Models with Data from Optical Coherence Tomography to Predict Future Visual Field Loss: The African Descent and Glaucoma Evaluation Study

Ophthalmology Science, 2022

Collaboration with Kellogg Eye Center and Shiley Eye Institute

M. Zhalechian, MP. Van Oyen, M. Lavieri, C. Andrew[†], J. Stein (M.D.), et al.

Working Papers

[8] From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement

Under Review

Collaboration with Michigan Medicine

A. Dean, M. Zhalechian, I. Attari, M. Garifullin[†], J. Pardo[†], MP. Van Oyen

- Implemented at Michigan Medicine since July 2023
- Selected for presentation at the Healthcare SIG Track, MSOM Conference, 2026
- Accepted at Michigan Medicine Quality Month Symposium, 2024

[9] **Does Early Pain Management Pay Off? Evidence on Utilization, Costs, and Surgeries Under Review**

Collaboration with Integrative Sports & Spine Pain Clinic in Southern California

B. Li, M. Zhalechian, C. Chen, N. Vaughn (M.D.), H. Lee (M.D.)

[10] **Learning Bundle Pricing of Reusable Products**

Under Review

A. Dean, M. Zhalechian, C. Shi

- Finalist, Service Science Best Student Paper Competition, 2024

[11] **Operations Management for Restorative Public Health: Grand Challenges in Reversing the Damage of the Opioid Crisis**

Major Revision at *Manufacturing & Service Operations Management*

C. Chen, J. Helm, B. Li, J. Mejia, M. Zhalechian (alphabetical authors)

- Selected for presentation at the Healthcare SIG Track, MSOM Conference, 2026

Selected Published Papers Prior to Ph.D.

Sustainable Design of a Closed-loop Location-routing-inventory Supply Chain Network Under Mixed Uncertainty

Transportation Research Part E, 2016

M. Zhalechian, R. Tavakkoli-Moghaddam, B. Zahiri, M. Mohammadi

Hub-and-spoke Network Design under Operational and Disruption Risks

Transportation Research Part E, 2017

M. Zhalechian, SA. Torabi, M. Mohammadi

A Self-adaptive Evolutionary Algorithm for a Fuzzy Multi-objective Hub Location Problem: An Integration of Responsiveness and Social Responsibility

Engineering Applications of Artificial Intelligence, 2017

M. Zhalechian, R. Tavakkoli-Moghaddam, Y. Rahimi

HONORS AND AWARDS

(* denotes honors or awards received by students)

2026 Finalist, MIF Paper Competition

2025 ODT Faculty Scholar for Research Excellence

2025 Kelley School of Business Research Award

2024* Finalist, Service Science Best Student Paper Competition

2022 Winner, Towner Prize for Distinguished Academic Achievement

Rackham Graduate School, University of Michigan

2022 Winner, Rackham Outstanding Graduate Student Instructor Award

Rackham Graduate School, University of Michigan

2022 Finalist, CHOM Best Paper Competition (POMS)

2021 Rackham Graduate Research Grant

Rackham Graduate School, University of Michigan

2021 Winner, Richard C. Wilson Best Student Paper Award

IOE Department, University of Michigan

2020 Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper
2020 Finalist, INFORMS Seth Bonder Scholarship for Applied OR in Health Services
2020 Winner, Joel and L. Brown Graduate Student Instructor of the Semester
 IOE Department, University of Michigan
2018–2021 Rackham Travel Grant Awards
2017 Graduate Student of the Year
 College of Engineering, University of Tehran

INVITED SEMINARS AND CONFERENCE PRESENTATIONS

From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement

- MSOM SIG Conference *Jul. 2026*

Interpretable Design of Multi-Agent Mobile Health Applications

- Workshop on AI & Analytics for Social Good, College Park, MD *Apr. 2026*
- INFORMS Healthcare Conference, Toronto, Canada *Jul. 2023*

Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy

- Wayne State University, Industrial & Systems Engineering Seminar Series *Nov. 2025*
- Carnegie Mellon University, Tepper School of Business *Feb. 2025*
- INFORMS Annual Meeting (2023, 2024, 2025); MSOM Conference (2025); POMS Annual Conference (2024); INFORMS Healthcare Conference (2023)

Data-Driven Learning and Resource Allocation in Service Operations Management

- Dartmouth College, Tuck School of Business *Jan. 2023*
- Indiana University, Kelley School of Business *Jan. 2023*
- Carnegie Mellon University, Tepper School of Business *Dec. 2022*
- Rice University, Jones Graduate School of Business *Dec. 2022*
- University of North Carolina at Chapel Hill, Kenan-Flagler Business School *Dec. 2022*
- University of Illinois Urbana-Champaign, Gies College of Business *Dec. 2022*
- Miami University, Herbert Business School *Dec. 2022*
- University of Illinois Chicago, College of Business Administration *Dec. 2022*
- Baruch College, Zicklin School of Business *Dec. 2022*

Personalized Hospital Admission Control: A Contextual Learning Approach

- INFORMS Annual Meeting (2019, 2020, 2021, 2022); POMS Annual Conference (2019, 2021, 2022); INFORMS Healthcare Conference (2019)

Online Advance Appointment Scheduling with Personalized Learning

- INFORMS Annual Meeting (2020, 2021)

Joint Learning and Optimization for Healthcare Operations and Medical Decision-Making

- INFORMS Annual Meeting, Seth Bonder Scholarship Session

Nov. 2020

A Distributionally Robust Capacity Planning Model for Optimizing Access Delay in Surgical Services under Limited Information

- INFORMS Healthcare Conference (2019); INFORMS Annual Meeting (2018)

TEACHING EXPERIENCE

Instructor, Indiana University, Bloomington, IN

Spring 2024, 2025 & 2026

Class: *Business Analytics and Modeling* (K353),

Three sections per year

Instructor, University of Michigan, Ann Arbor, MI

Fall 2020

Class: *Economic Decision Making* (IOE 201)

Recitation Instructor, University of Michigan, Ann Arbor, MI

Winter 2020

Class: *Introduction to Markov Processes* (IOE 316)

Teaching Assistant, University of Michigan, Ann Arbor, MI

- Practicum in Hospital Systems (IOE 481)

Fall 2021 – Winter 2022

- Service Operations Management (IOE 419)

Winter 2021

- Probability and Statistics for Eng. (IOE 265)

Fall 2018, Winter 2019, Fall 2019

ADVISING AND MENTORING

PhD Student Committees

- Ali Kaan Kurbanzade, Kelley School of Business, 2025–present
- Iman Attari, Kelley School of Business, 2023–2025
Placement: University of Cincinnati (Lindner College of Business)

Master's and Undergraduate Mentoring

- Faculty Mentor for Master's Students, Faculty Assistance in Data Science (FADS) Program, Kelley School of Business
- Faculty Mentor for Undergraduate Students, Center of Excellence for Women & Technology, Indiana University

ACADEMIC SERVICE

Service to the Profession

- *Referee*: Management Science, Operations Research, Manufacturing & Service Operations Management, Production and Operations Management, Information Systems Research, European Journal of Operational Research, Omega, Health Care Management Science
- *Judge*: Public Sector OR Best Paper Competition, INFORMS, 2023

Session Organizer & Chair for Conferences

2026 INFORMS Annual Meeting (AI in Healthcare); INFORMS Healthcare Conference (AI for Healthcare Oversight and Operations & AI and Analytics for Decision-Making in Healthcare)

2025 INFORMS Annual Meeting (Data-driven Healthcare Analytics)

2024 POMS Annual Conference (Policies and Regulation in Healthcare)

2021 INFORMS Annual Meeting (Sequential Decision Making and Machine Learning in Healthcare); POMS Annual Conference (Healthcare Operations Management)

2020 INFORMS Annual Meeting (Predictive Analytics & Sequential Decision-Making in Healthcare)

Department and School Service

- PhD Recruiting Committee, Operations and Decision Technologies, Kelley School of Business, 2025
- Faculty Recruiting Committee, Operations and Decision Technologies, Kelley School of Business, 2024
- Faculty Recruiting Committee (Graduate Student Committee), Industrial and Operations Engineering, University of Michigan, 2021
- Chair, Anti-Racism and DEI Committee, INFORMS Student Chapter, University of Michigan, 2020–2021
- Chair, Professional Development Committee, INFORMS Student Chapter, University of Michigan, 2019–2020